Replay-Aware CVRP Aggregate Report

Overview

- Run count: 20.
- Condition count: 4.
- Best mean transfer gap: cvrp_no_replay.

Condition Comparison

Condition	Mean Transfer Gap	Mean CVRPLIB Gap	Mean Synthetic Gap	Mean Accepted Novelty	Mean Complexity	Mean Adaptation Efficiency
cvrp_failure_replay	0.137238	0.238397	0.010788	0.596004	0.67	0.020111
cvrp_failure_replay_compression	0.140394	0.243836	0.011093	0.511511	0.63	0.018193
cvrp_no_replay	0.136873	0.235779	0.01324	0.711362	0.7	0.013129
cvrp_random_replay	0.139859	0.246032	0.007144	0.766976	0.63	0.013062

Bootstrap Confidence Intervals

cvrp_failure_replay

- Transfer gap 95% CI: 0.133831 to 0.14071.
- CVRPLIB gap 95% CI: 0.233061 to 0.243465.
- Synthetic gap 95% CI: 0.005491 to 0.017588.
- Mean accepted novelty 95% CI: 0.418211 to 0.764053.

cvrp_failure_replay_compression

- Transfer gap 95% CI: 0.135442 to 0.145871.
- CVRPLIB gap 95% CI: 0.236307 to 0.250893.
- Synthetic gap 95% CI: 0.005107 to 0.018784.
- Mean accepted novelty 95% CI: 0.328095 to 0.68867.

cvrp_no_replay

- Transfer gap 95% CI: 0.133036 to 0.140191.
- CVRPLIB gap 95% CI: 0.229341 to 0.241658.
- Synthetic gap 95% CI: 0.007953 to 0.019903.
- Mean accepted novelty 95% CI: 0.570133 to 0.831854.

cvrp_random_replay

- Transfer gap 95% CI: 0.13685 to 0.143267.
- CVRPLIB gap 95% CI: 0.240936 to 0.251195.

- Synthetic gap 95% CI: 0.005376 to 0.009667.
- Mean accepted novelty 95% CI: 0.63726 to 0.863645.

Paired Comparisons

Negative delta favors the candidate for transfer gap, benchmark gap, synthetic gap, novelty, and complexity. Positive delta favors the candidate for adaptation efficiency.

Comparison	Paired Runs	Transfer Delta	Transfer 95% CI	CVRPLIB Delta	CVRPLIB 95% CI	Novelty Delta	Complexity Delta	Adaptation Delta
cvrp_random_replay vs cvrp_no_replay	20	0.002987	-0.001817 to 0.008152	0.010253	0.002587 to 0.018356	0.055614	-0.07	-6.7e-05
cvrp_failure_replay vs cvrp_no_replay	20	0.000365	-0.004065 to 0.005159	0.002618	-0.002573 to 0.007785	-0.115358	-0.03	0.006981
cvrp_failure_replay vs cvrp_random_replay	20	-0.002622	-0.006831 to 0.001333	-0.007634	-0.014563 to -0.000625	-0.170972	0.04	0.007048
cvrp_failure_replay_compression vs cvrp_failure_replay	20	0.003157	-0.001624 to 0.007804	0.005438	-0.002502 to 0.012963	-0.084493	-0.04	-0.001918
cvrp_failure_replay_compression vs cvrp_no_replay	20	0.003522	-0.002561 to 0.010426	0.008056	-0.000843 to 0.01684	-0.199851	-0.07	0.005064

Paired Notes

- cvrp_random_replay vs cvrp_no_replay used 20 paired runs.
- Mean transfer-gap delta: 0.002987 with 95% CI -0.001817 to 0.008152; candidate better on 50.0% of paired runs.
- Mean CVRPLIB-gap delta: 0.010253 with 95% CI 0.002587 to 0.018356.
- cvrp_failure_replay vs cvrp_no_replay used 20 paired runs.
- Mean transfer-gap delta: 0.000365 with 95% CI -0.004065 to 0.005159; candidate better on 50.0% of paired runs.
- Mean CVRPLIB-gap delta: 0.002618 with 95% CI -0.002573 to 0.007785.
- cvrp_failure_replay vs cvrp_random_replay used 20 paired runs.
- Mean transfer-gap delta: -0.002622 with 95% CI -0.006831 to 0.001333; candidate better on 65.0% of paired runs.
- Mean CVRPLIB-gap delta: -0.007634 with 95% CI -0.014563 to -0.000625.
- cvrp_failure_replay_compression vs cvrp_failure_replay used 20 paired runs.
- Mean transfer-gap delta: 0.003157 with 95% CI -0.001624 to 0.007804; candidate better on 35.0% of paired runs.
- Mean CVRPLIB-gap delta: 0.005438 with 95% CI -0.002502 to 0.012963.
- cvrp_failure_replay_compression vs cvrp_no_replay used 20 paired runs.
- Mean transfer-gap delta: 0.003522 with 95% CI -0.002561 to 0.010426; candidate better on 45.0% of paired runs.
- Mean CVRPLIB-gap delta: 0.008056 with 95% CI -0.000843 to 0.01684.